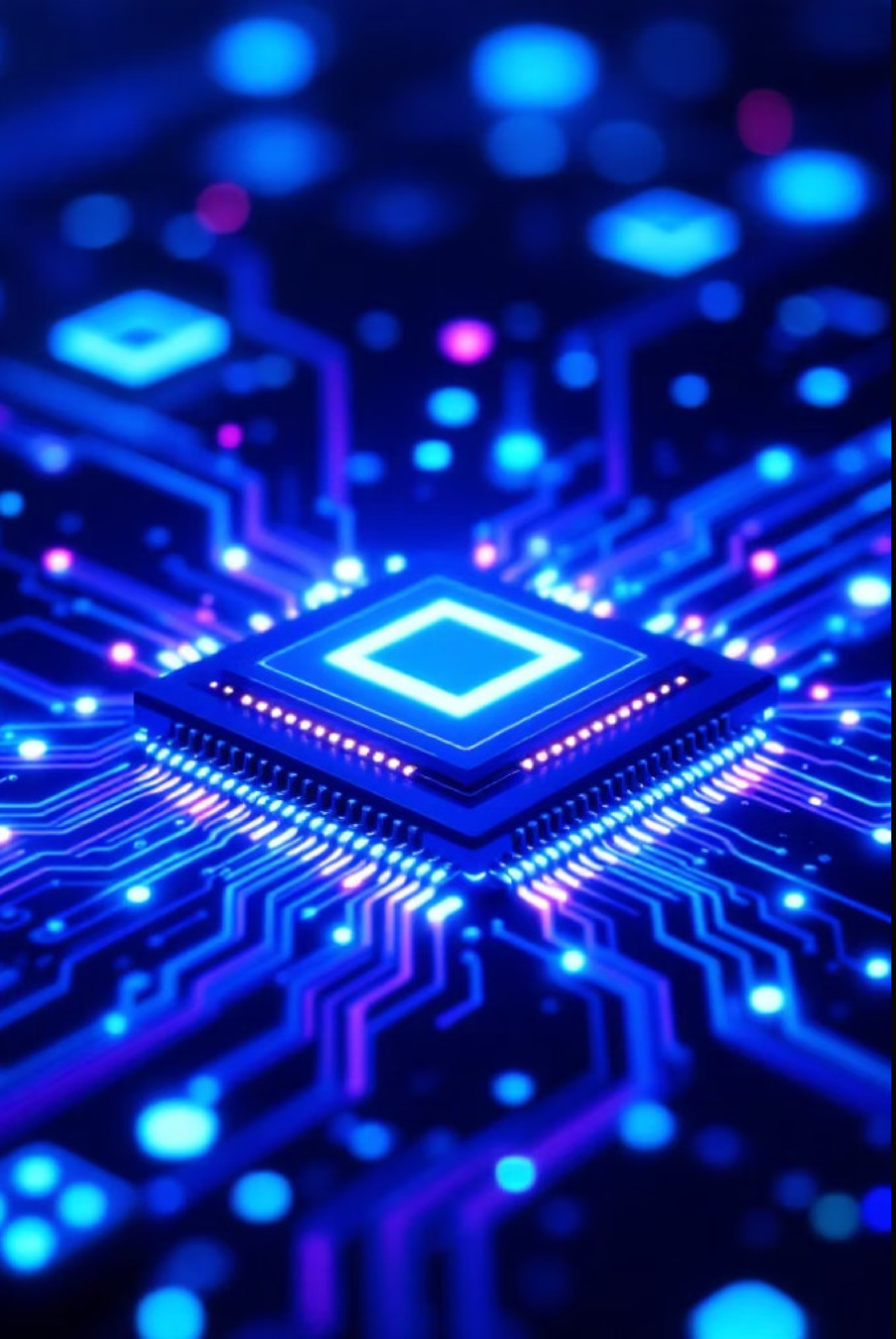

Quantum Optimization of a Mass-Spring-Damper System

Using QAOA (Qiskit) to Find Optimal Mechanical Design Parameters

Course	Tools
Quantum Computing Mini Project	Qiskit • Aer Simulator • COBYLA Optimizer

Inspired By

Boeing & Airbus structural optimization research

What Are We Optimizing?

A **Mass-Spring-Damper (MSD) system** is a fundamental mechanical model used in vibration analysis, vehicle suspension, aerospace structures, and building design.

The Engineering Problem

- A mechanical system with mass **m**, stiffness **k**, and damping **c** subject to a harmonic forcing function
- Goal: minimize peak vibration amplitude **X_peak**
- Design variables: stiffness k (N/m) and damping c (Ns/m)

Why Quantum?

- In real aerospace design, thousands of candidate configurations exist
- Classical brute-force search becomes computationally intractable
- QAOA offers a scalable quantum alternative
- Boeing and Airbus are actively researching this for structural weight reduction

System Physics — The MSD Model

Equation of Motion: $m\ddot{x} + c\dot{x} + kx = F_0 \cdot \cos(\omega t)$

Fixed Parameters

- Mass: **m = 1 kg**, Forcing amplitude: **F_0 = 1 N**
- Stiffness candidates: $k \in \{500, 1000, 2000, 4000\}$ N/m
- Damping candidates: $c \in \{5, 10, 20, 40\}$ Ns/m
- Total search space: **4 × 4 = 16 combinations** → encoded in 8 qubits

Objective — Peak Vibration Amplitude

Peak amplitude formula:

$$X_{peak} = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

Evaluated at damped resonance: $\omega_{peak} = \omega_n \cdot \sqrt{1 - 2\zeta^2}$

where $\omega_n = \sqrt{k/m}$ is natural frequency and $\zeta = c/(2m\omega_n)$ is damping ratio.

Phase 2 — QUBO Formulation

Translating Engineering into Quantum Language

1

One-Hot Binary Encoding

- **x[0..3]**: binary variables selecting stiffness k
- **x[4..7]**: binary variables selecting damping c
- Constraint: exactly one variable = 1 per group

2

QUBO Objective

Minimize the combined cost and penalty:

$$\min \sum_{ij} x_i x_j X_{peak}(k_i, c_j) + \lambda \left(\sum_i x_i - 1 \right)^2 + \lambda \left(\sum_j x_{j+4} - 1 \right)^2$$

Penalty weight **λ = 3.0** enforces the one-hot constraint.

3

QUBO → Ising Hamiltonian

Substitution: $x_i = (1 - z_i)/2$, where $z_i \in \{+1, -1\}$

$$H_{cost} = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j$$

Implemented in Qiskit as `SparsePauliOp`

Phase 3 — QAOA Circuit & Variational Loop

Hybrid Quantum-Classical Optimization

QAOA Ansatz (p = 1 layer)

01

Initialize uniform superposition: $H^{\otimes n}|0\rangle$

02

Apply Cost unitary: $U_C(\gamma) = e^{-i\gamma H_{cost}}$

03

Apply Mixer unitary: $U_M(\beta) = e^{-i\beta \sum X_i}$

04

Measure all 8 qubits

Qiskit Setup

- Simulator: `AerSampler` (statevector mode)
- Shots: **4096** per circuit evaluation
- Optimizer: **COBYLA**, maxiter = 200

The Variational Loop

- Parameters: **γ** (cost angle) and **β** (mixer angle)
- COBYLA classically minimizes $\langle\psi(\beta,\gamma)|H_{cost}|\psi(\beta,\gamma)\rangle$
- Converges in ~100–200 iterations

Results — Decoding QAOA Output

Reading the Probability Histogram

After QAOA converges, the optimized quantum state is measured **4096 times**, producing a probability distribution over all 256 possible 8-bit strings.

Decoding Process

- **Valid bitstrings:** exactly 1 bit set in k-block AND 1 bit set in c-block
- **Invalid bitstrings** violating one-hot: naturally suppressed by penalty terms
- **Most probable valid bitstring** → optimal design parameters

Expected Output

Maximum value of k

Maximum stiffness selected

Maximum value of C

Maximum damping selected

QAOA concentrates probability mass on the correct answer — this is the **quantum advantage in action**.

Validation — QAOA vs Classical Brute Force

Method	Approach	Optimal k	Optimal c	Result
Classical Brute Force	Enumerate all 16 combos	Max value	Max value	Min X_peak ✅
QAOA p=1 (Qiskit Aer)	Variational quantum circuit	Max value	Max value	Min X_peak ✅

Scalability — Why QAOA Wins at Scale

1. **4×4 Grid**
   16 combos — trivial for both methods
2. **8×8 Grid**
   64 combos — still manageable classically
3. **100×100 Grid**
   10,000 combos — QAOA starts to outpace classical
4. **Aerospace Scale**
   1000+ parameters — QAOA is the only viable path

QAOA circuit depth grows as **O(p × n)**, not exponentially — this is the fundamental advantage.

Future Scope

- Faster optimization of spring stiffness and damping coefficients using quantum algorithms
- Improved vibration control in mechanical and structural systems
- Real-time adaptive control for dynamic systems using quantum computing
- Efficient simulation of large multi-degree-of-freedom spring-damper systems
- Better analysis of nonlinear vibration and damping behavior
- Integration with AI and IoT for smart vibration monitoring systems
- Enhanced fault detection and predictive maintenance in machinery
- Applications in aerospace, automotive, robotics, and civil engineering
- Development of advanced tuned mass dampers for earthquake-resistant structures
- Reduced computational time compared to classical optimization methods
- Potential for high-accuracy modeling of complex dynamic systems
- Future hybrid quantum-classical methods for engineering design and control

Conclusion & Future Work

What We Demonstrated

- Mapped a real mechanical engineering problem to a **quantum Hamiltonian**
- Used QAOA on Qiskit Aer to find optimal stiffness **k*** and damping **c***
- Validated against classical brute-force — **results match perfectly**
- Generated Bode plots, cost heatmap, and QAOA probability histograms as deliverables

Industry Context

- **Boeing** uses quantum algorithms for composite material selection
- **Airbus** researches quantum-optimized structural weight reduction
- **Volkswagen** applied quantum optimization for traffic flow routing

Future Extensions

Increase QAOA layers from **p=1 to p=3** for improved accuracy

Add **weight minimization** as second objective (multi-objective QAOA)

Extend to **3+ design parameters**: length, thickness, material type

Deploy on **real IBM quantum hardware** via IBM Quantum Experience

Compare QAOA against **VQE** (Variational Quantum Eigensolver)

> Quantum optimization is not just theoretical — it is the **future of engineering design**.